Microsoft Power BI
Part 2

Report

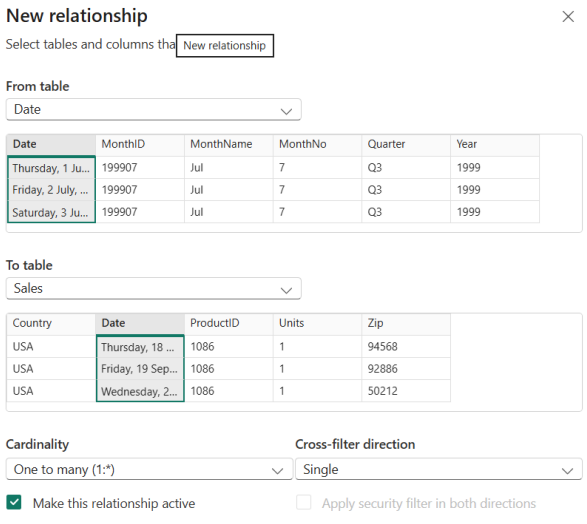
Microsoft

Power BI

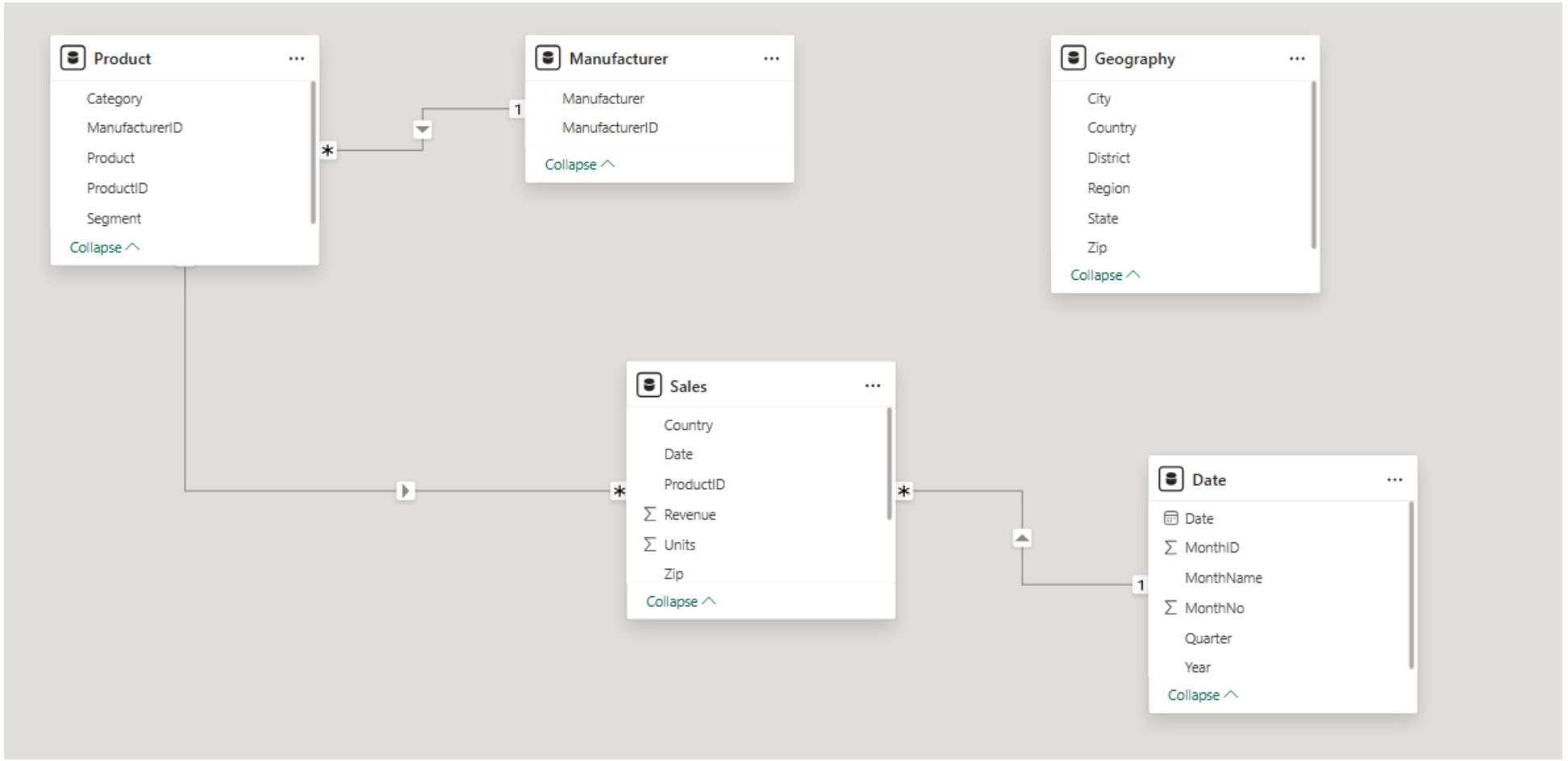
Data Model–Model View

Click on the Model Views on the left panel of Power BI Desktop.

1. You will see the tables you have imported along with some Relationships..
   –Relationship is created between Sales and Product tables using ProductID column.
   –Relationship is created between Product and Manufacturer tables using ManufacturerID column.
2. Need to create relationship between Sales and to other dimensional tables.
3. Relationships in one of the tables needs to have unique values. (Power BI desktop supports 1 to many or 1 to 1 relationships between the tables.
3. Create relationship between the sales and date table to build time intelligence measures.
   – Drag the Date field from the Sales table to the Date field in the Date table.



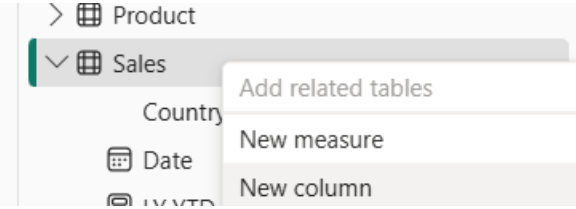
Data Model–Model View



Data Model–Model View

Set up the correct relationship between the Geography table and the Sales table.
- Create a new column in both the Geography table and the Sales table that combines “Zip” and “Country” to have a unique field.

1) Go to the report View.
2) In the data section, click on the ellipsis next to Sales table.
2) Select “New Column” and the formula editor appear to help create this new column.



3) Combine or concatenate the Zip and Country columns into a new column called ZipCountry separated by a comma.

ZipCountry = Sales[Zip]&","&Sales[Country]

```
1  ZipCountry = Sales[Zip]&","&Sales[Country]
```

4) Click in the check mark on the left side of the formula editor. A new column ZipCountry in Sales table is created.

Data Model-Model View

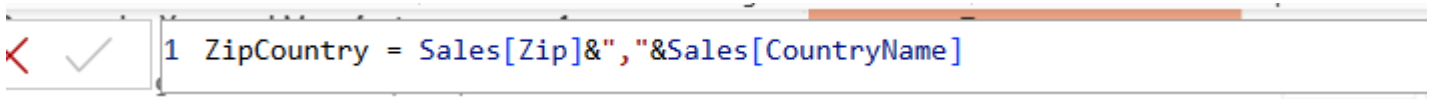
-A new column also can be created by selecting the table and then clicking on

Modeling -> New Column from the ribbon.

1.Create a “ZipCountry” column in the Geography table.

2. From Data section, select Geography table and from the ribbon select Modeling -> New Column Formula bar appears.

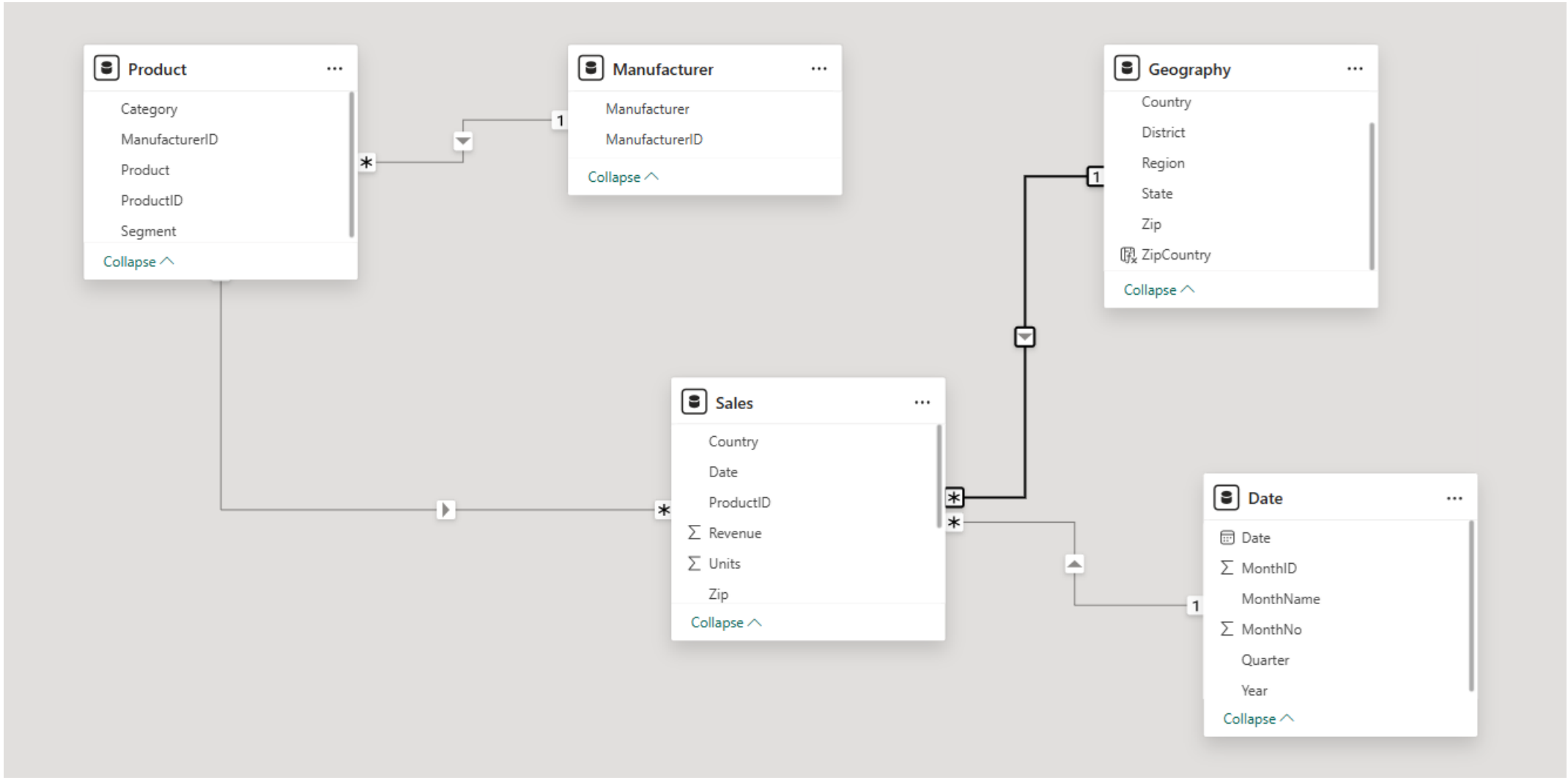
3. Enter the following DAX expression in the formula bar:
ZipCountry = Geography[Zip]&","&Geography[Country]



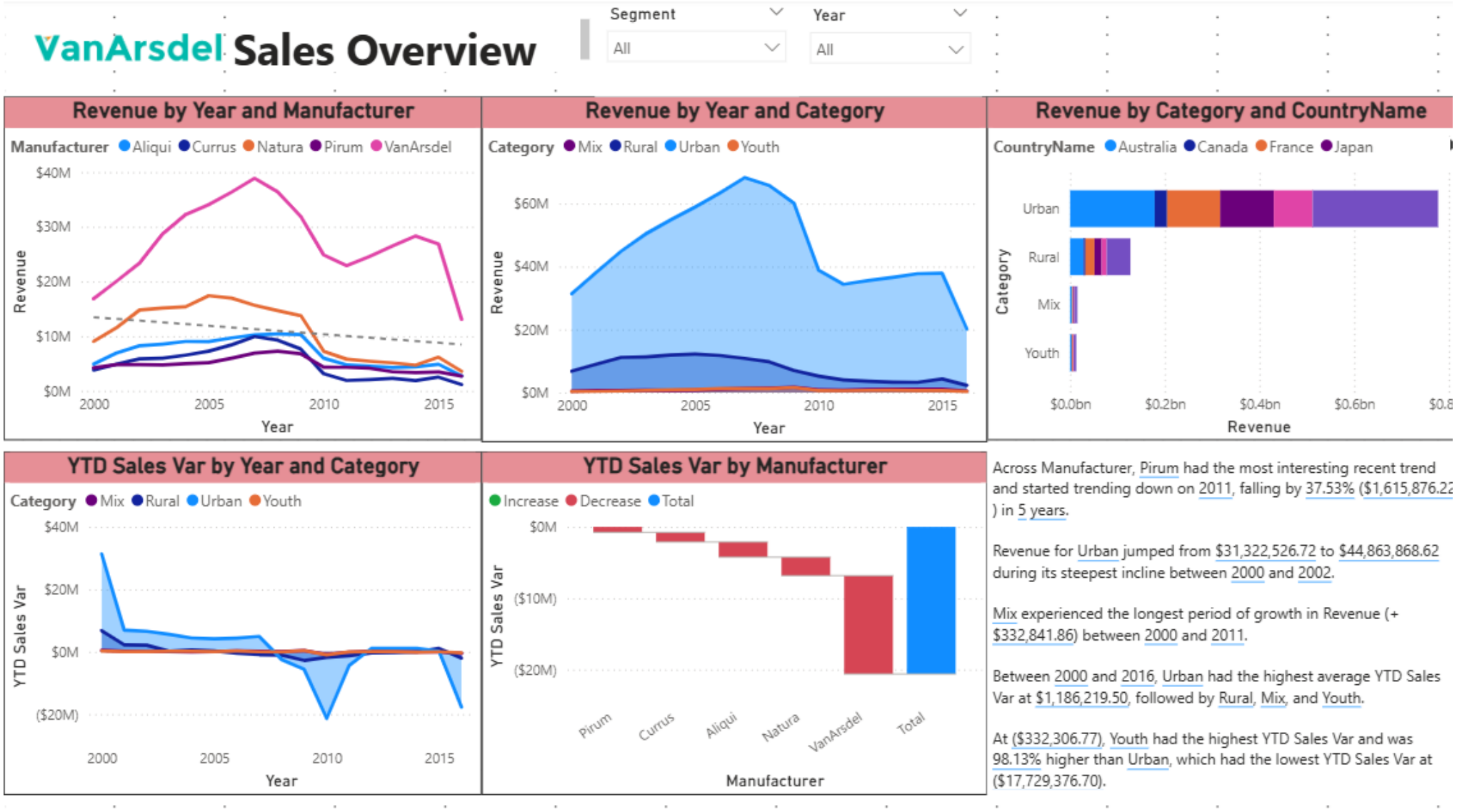
4. A new column ZipCountry is created in Geography table.

5. Then setup the relationship between the two tables using the newly created “ZipCountry” columns between Sales And Geography table.

Data Model–Model View



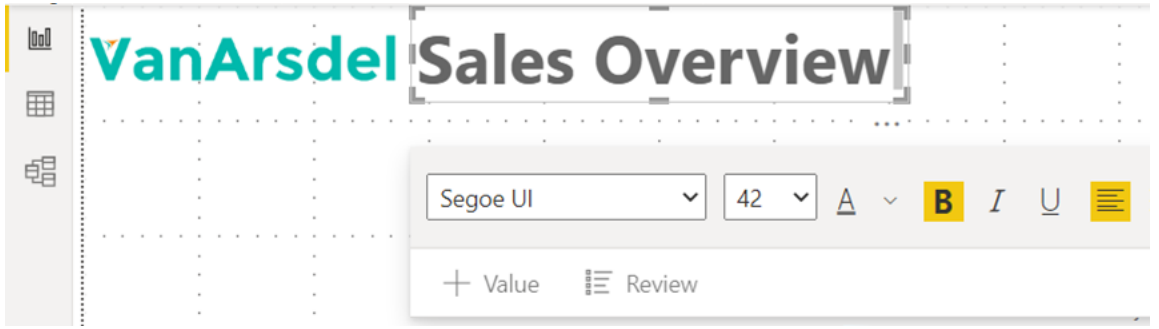
Report View–Sales Overview



Power BI Desktop – Add Image and Text

You need the company logo in most reports.

1. Click on the white space on the canvas.
2. From the ribbon, select Insert -> Image.
3. Navigate to image file and select Logo.png. Click Open.
4. Logo will appear on the report.
5. Resize and move the logo to the top left of the report.



Next let us add a title to this report. To do so click anywhere on the white canvas.

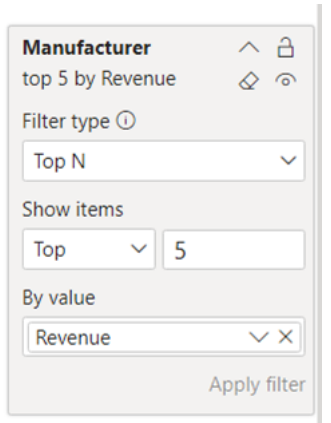
1. From the ribbon, select Insert -> Text box. Text box area is available in the canvas.
2. Enter the title “Sales Overview”.
3. Highlight Sales section of the header and change the font to Segoe UI, font size to 42 and make the text bold.

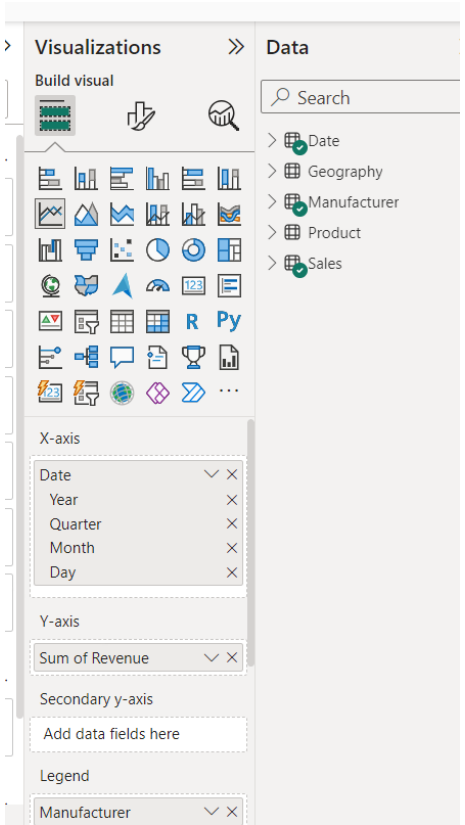
Power BI Desktop – Line graph

1. From Visualizations section select Line Chart .
2. From the Data section expand Sales table and drag and drop Revenue to Y-axis.
3. From the Data section expand Date table and drag and drop Date to X-axis
4. From the Data section, expand Manufacturer table and drag and drop Manufacturer to the Legend of the visual
5. From Filters section, scroll to Visual level filters.
6. Expand Manufacturer and from the Filter Type drop down select Top N.
7. Enter 5 next to Top since we are comparing the Top 4 competitors
8. From the Data section, expand Sales table and drag and drop Revenue to By Value.
   We are doing this to get the Top 5 competitors based on revenue.
9. Select Apply Filter

It will be useful to display Units sold along with Revenue information on the tooltip.

10. From the Data section, expand Sales and drag Units to Tooltips

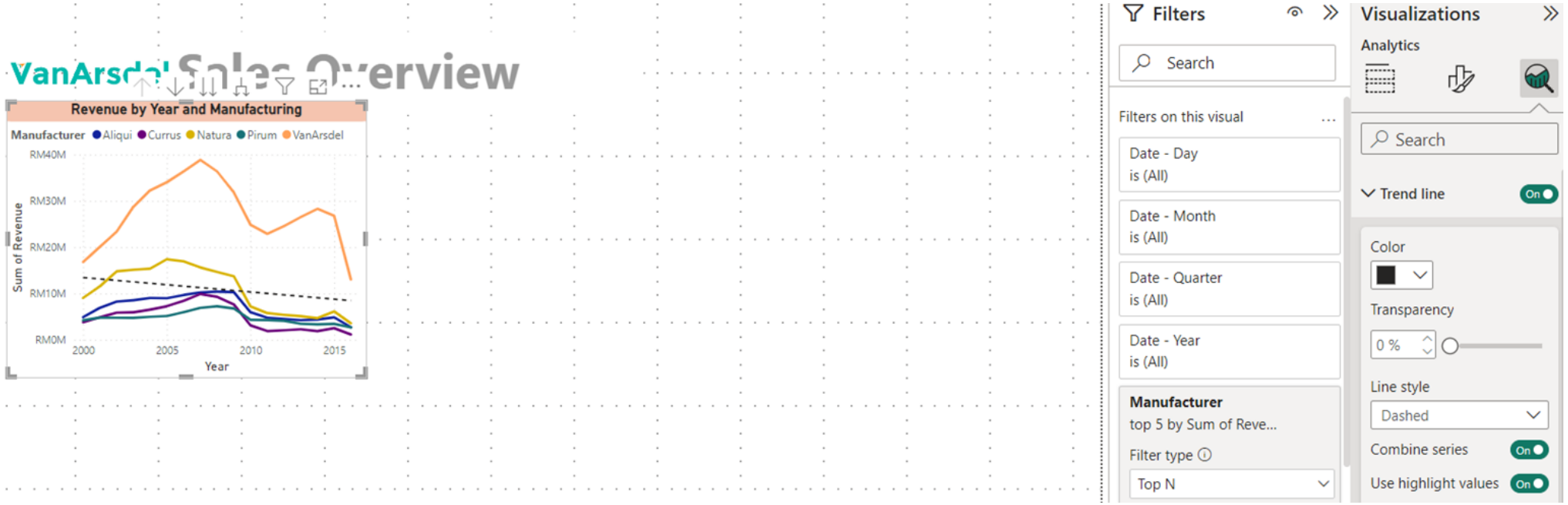




Power BI Desktop – Trend Line

Power BI Desktop provides capability to perform analytics on visuals. It's useful to know the revenue trend. Let's add it to the Line chart.

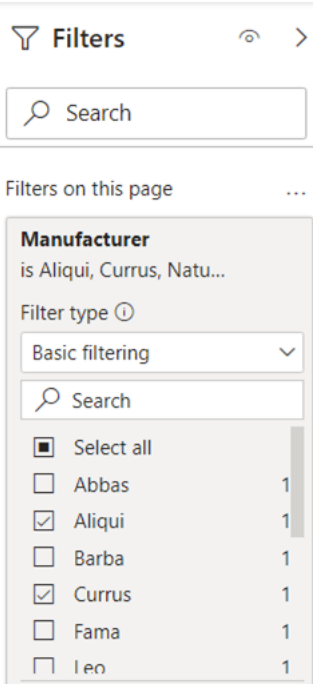
1. With the Line chart selected, from the Visualization click on the Analytics icon.
2. Notice there is options to add Trend, Constant, Min, Max, Average, Median and Percentile Line.
3. Turn On Trend line



Power BI Desktop – Trend Line

Earlier we used Top N feature to filter the column chart down to Top competitors.
Suppose we want to apply this filter to all the visuals in a page,

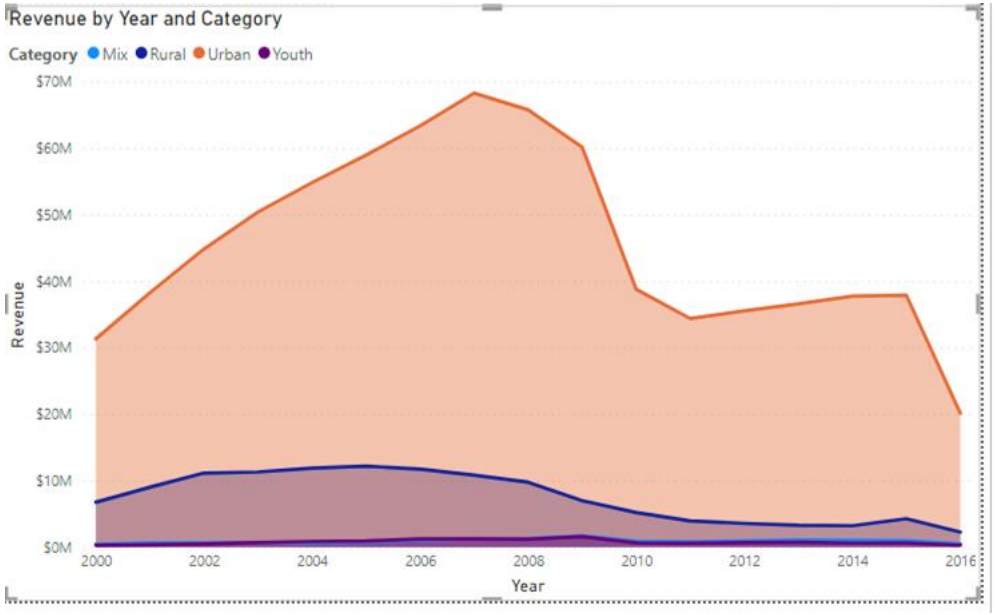
1. Click on the white space on the canvas
2. In Filters section , add the Manufacturer column to Filters on this page
3. Select the manufacturers Aliqui, Currus, Natura, Pirum and VanArsdel in the filter

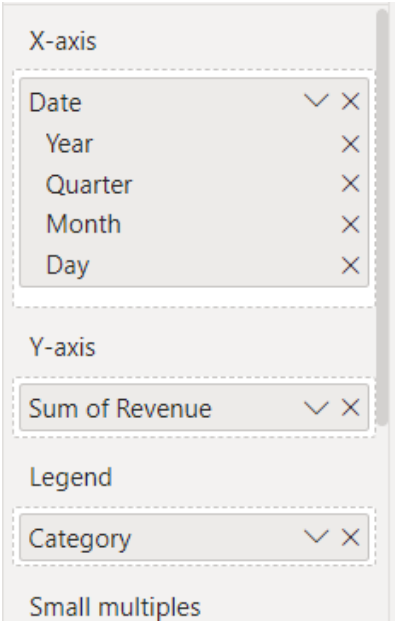


Power BI Desktop – Area Graph

Revenue by Category

1. Click in the report canvas.
2. Select the Stacked area chart visual from the Visualizations section.
3. From the Data section, expand Date table and drag and drop Date field to X-axis.
4. From the Data section, expand Product table and drag and drop Category field to Legend.
5. From the Data section, expand Sales table and drag and drop Revenue field to Y-axis.
6. Resize the chart as needed.

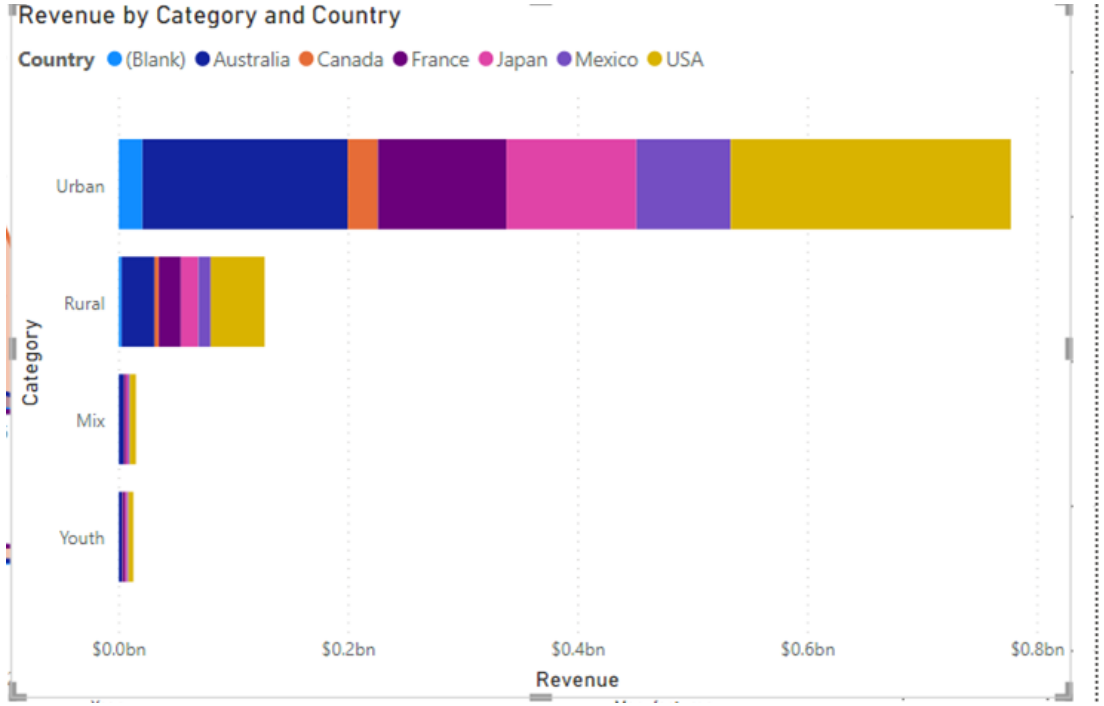


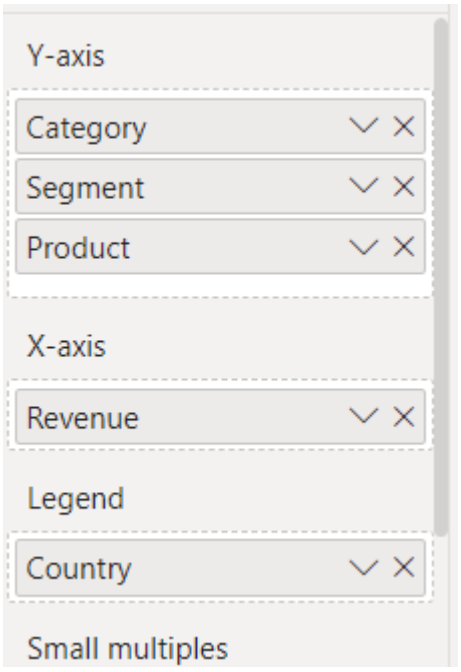


Power BI Desktop – Hierarchy Stack Bar Chart

This chart provides Revenue by Country by Product hierarchy

1. Select the Stacked bar chart visual from the Visualizations section.
2. From the Data section, expand Product table and drag and drop Category ,Segment and Product field to Y-axis
3. From the Data section, expand Geography table and drag and drop Country field to Legend.
4. From the Data section, expand Sales table and drag and drop Revenue field to X-axis.
5. Feel free to drill up and down the Product hierarchy and analyze the revenue





DAX (Data Analysis Expressions)

- DAX (Data Analysis Expressions) is a formula expression language include functions, operators,and values to perform advanced calculations and queries on data in related tables and columns in tabular data models.

- By using Data Analysis Expressions (DAX), you can add three types of calculations to your data model:

i. Calculated tables
ii. Calculated columns
iii. Measures

Power BI Desktop – DAX (Year To Date)

You want the ability to compare current Year to Date sales with the previous year and see if you are doing better or worse. To achieve such calculations in Power BI desktop you need to create a Measure using a DAX functions that understand Time calculations such as “Year to Date”

1.To create a new Measure, from Data section click on the ellipsis next to Sales table.
2. Select New Measure. Notice a formula bar appears.
3. In the formula bar enter the following DAX Expression

```
YTD Revenue = TOTALYTD(sum(Sales[Revenue]),'Date'[Date])
```

4.Click on the check mark next to the formula bar.
5.You will see the YTD Revenue under Sales table with a special icon which indicates it is a measure (calculator).

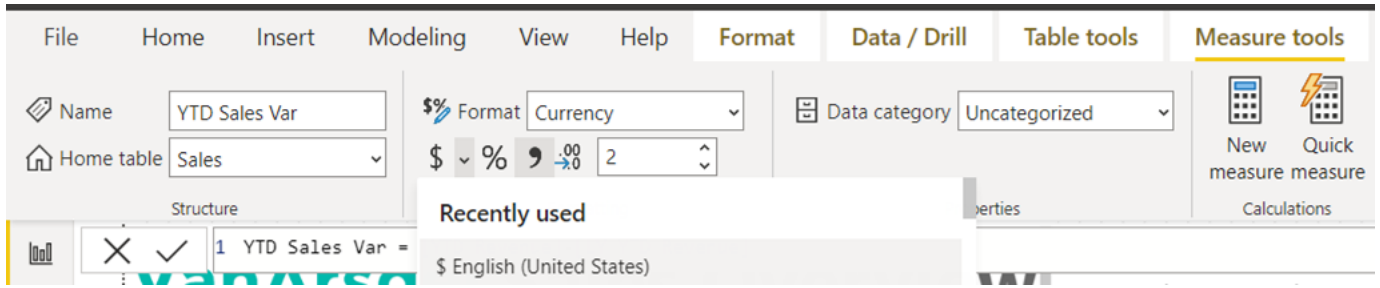
Similarly, let’s create Last Year to Date Revenue measure and Year To date Sales variable

```
LY YTD Revenue = CALCULATE([YTD Revenue],SAMEPERIODLASTYEAR('Date'[Date]))
YTD Sales Var = [YTD Revenue]-[LY YTD Revenue]
```

Power BI Desktop – Formatting

- Let's format the newly created measures.

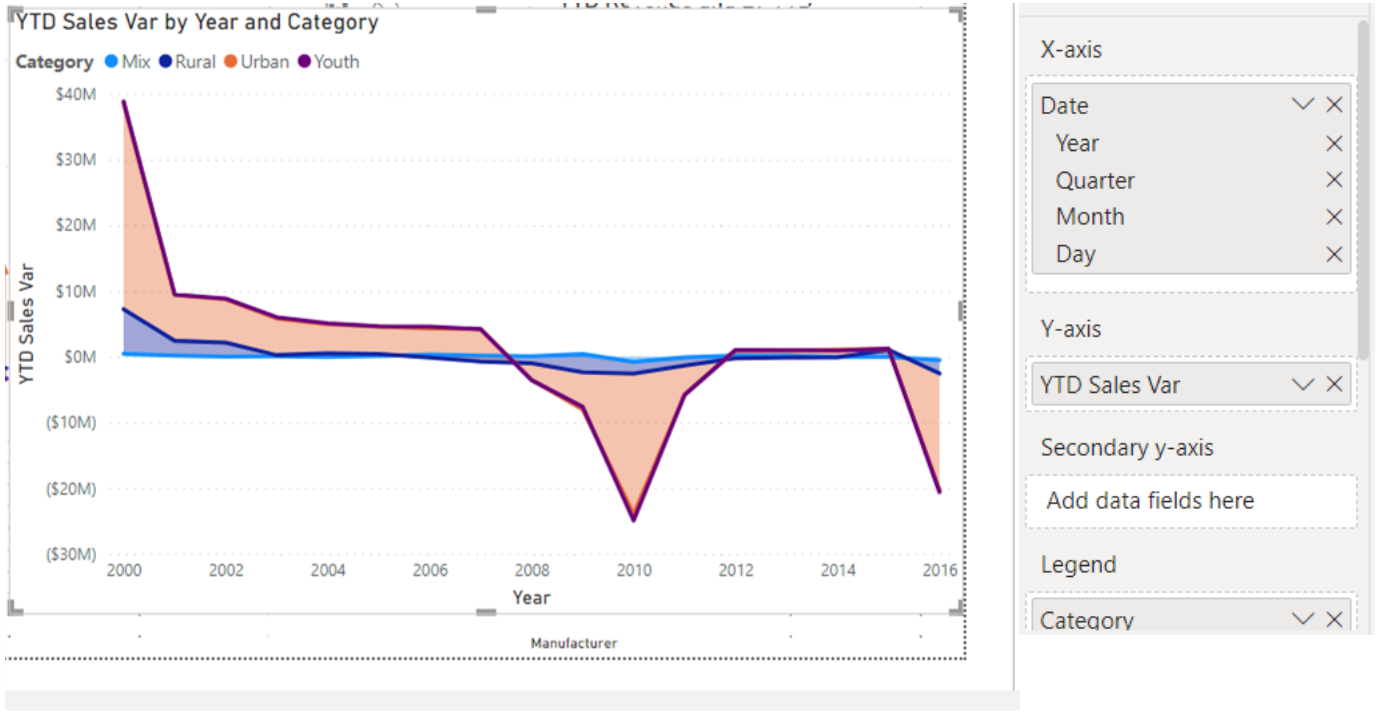
1. From the Data section, expand Sales table and select YTD Revenue
2. From the ribbon, select Measure Tools -> Format -> Currency -> $ English (United States).
3. Similarly, format LY YTD Revenue and YTD Sales Var measures



Power BI Desktop

This chart provides sales trend by product category over time

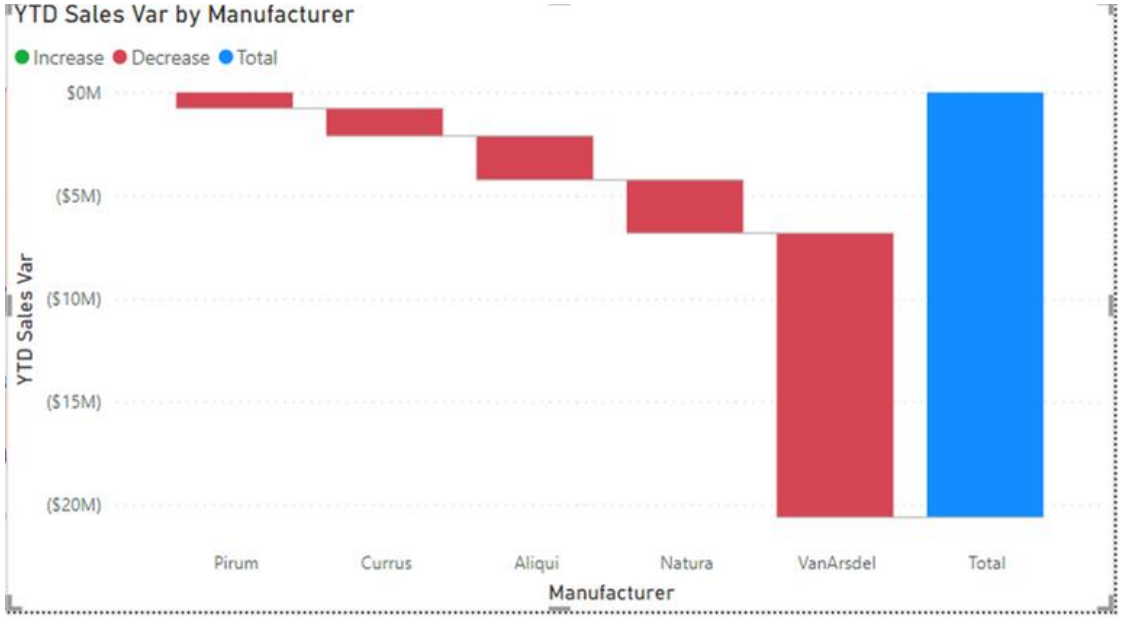
1. Select the Stacked bar chart visual from the Visualizations section.
2. From the Data section, expand Date table and drag and drop date field to X-Axis.
3. From the Data section, expand Product table and drag and drop Category field to Legend.
4. From the Data section, expand Sales table and drag and drop YTD Sales Var field to Y-Axis

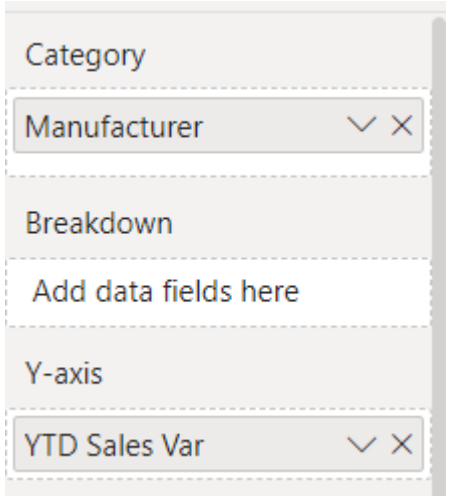


Power BI Desktop-Waterfall chart

This chart provides the analysis to find out how all the Manufacturers have performed overtime.

1. Select the Waterfall chart visual from Visualizations.
2. From the Data section, expand Manufacturer table and drag and drop Manufacturer field to Category section.
3. From the Data section, expand Sales table and drag and drop YTD Sales Var field to Y-axis



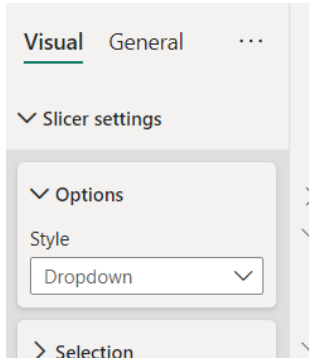


Power BI Desktop-Slicer

A slicer provides the ability for end users to visually see the key filters applied to the entire page and the flexibility to select/deselect values easily.

Date slicer

1.Click on the slicer icon in Visualizations pane.
2. From the Data section, expand Date table and select Year column and drag and drop to Field section
3. Go to Format and select Slicer Settings . Select Dropdown to change the slicer from a List to a Dropdown. Resize the slicer as needed.
4. Move the slicer to the top right corner of the canvas.



Let's add a slicer which will help analyze the data by Product Segment.

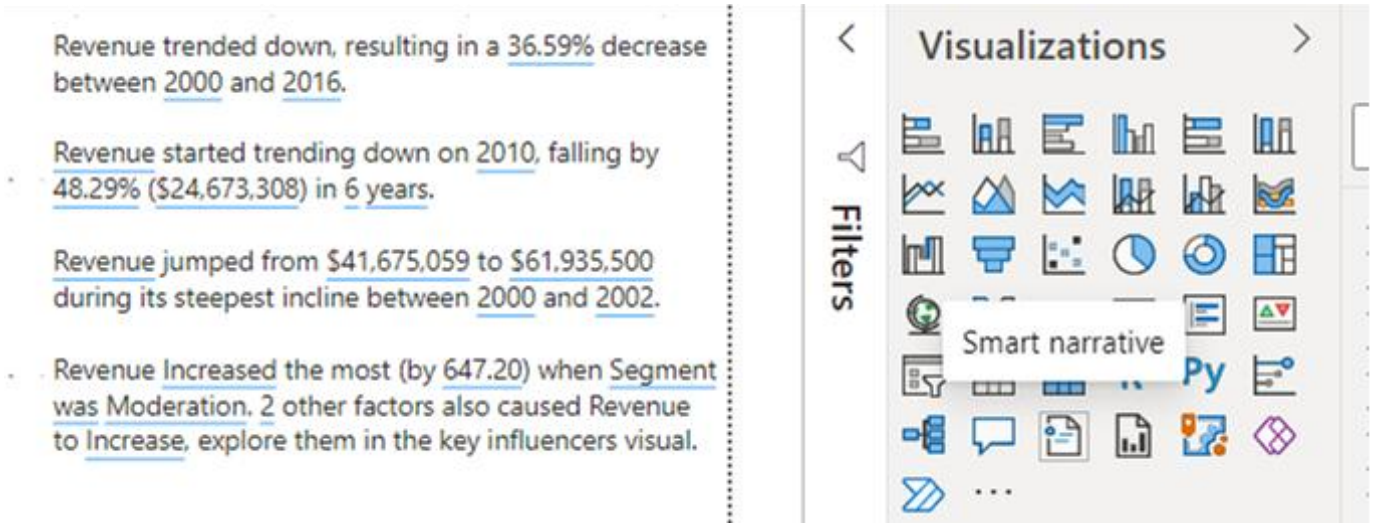
1. Click on the slicer icon in Visualizations pane.
2. From Data section, expand Product table and select Segment column and drag and drop to Field section .
3. Change the slicer from a List to a Dropdown.

Power BI Desktop–Smart Narative

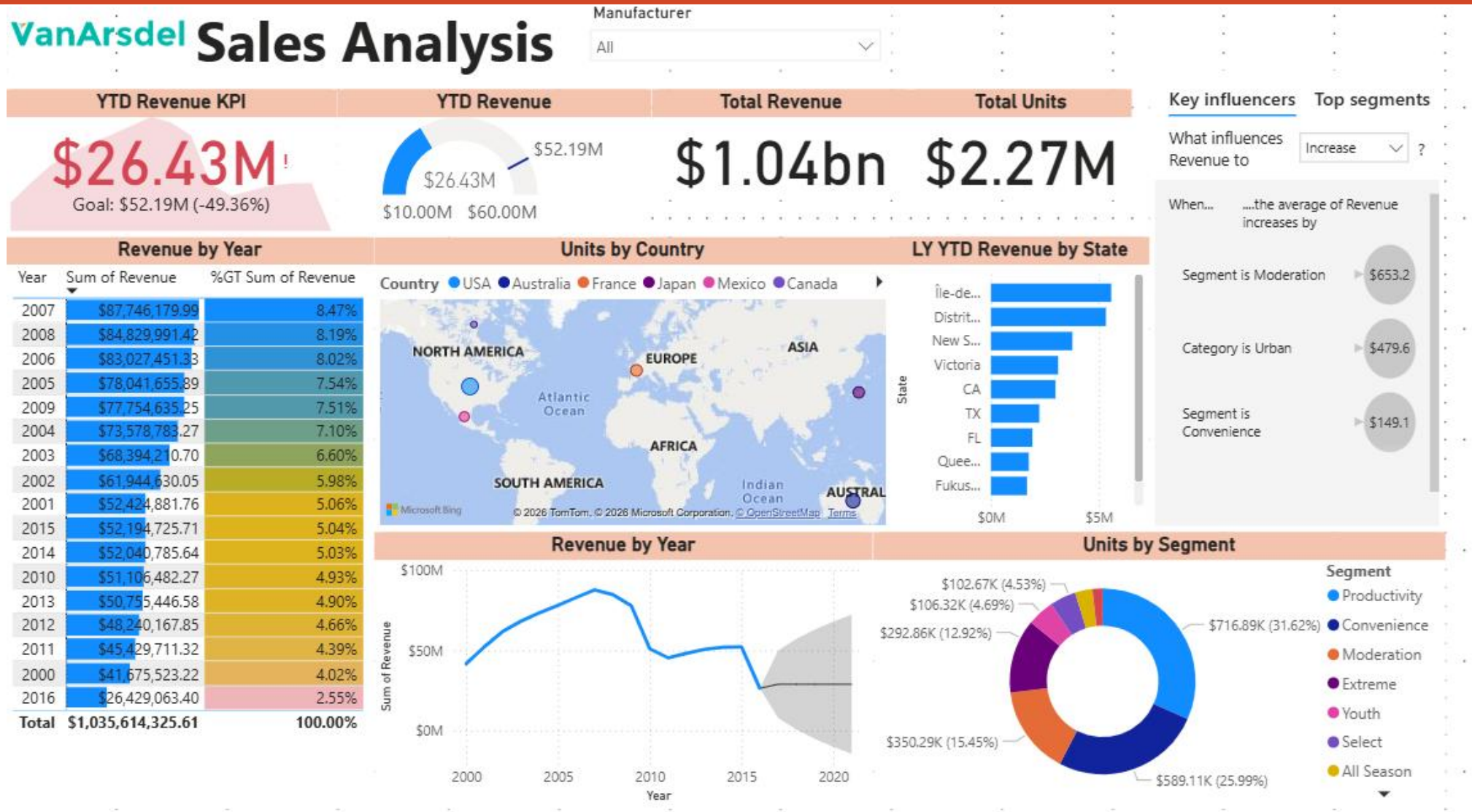
The smart narrative visualization helps you quickly summarize visuals and reports. It provides relevant innovative insights that you can customize

1. In the **Visualizations** pane, select the **Smart narrative** icon to automatically generate a summary.

1. You see a narrative that's based on all of the visuals on the page. For example, in the sample file, smart narratives can automatically generate a summary of the report's visuals that address revenue,



Report View –Sales Analysis(Page 2)



Power BI Desktop-KPI

Let's add a KPI visual that compares this year's revenue versus last year revenue

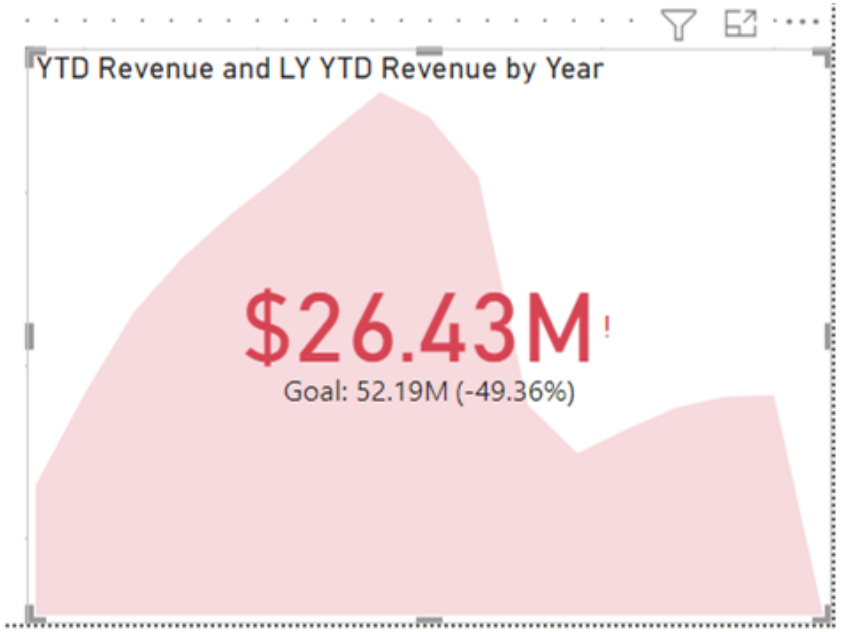
1.Click on the whitespace in the canvas.

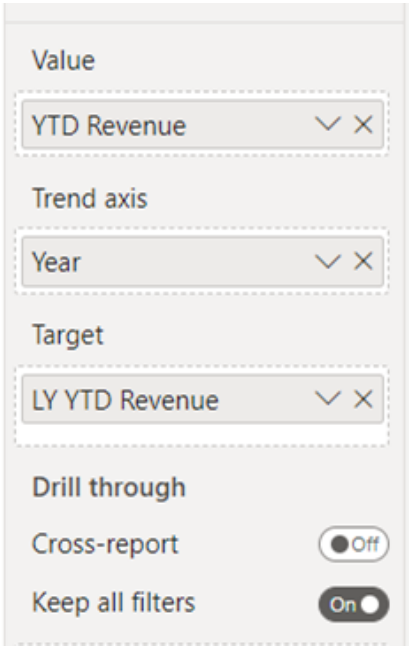
2. Click on the KPI visual in Visualizations section.

3. From the Data section, expand Date table and drag and drop Year field to Trend axis section.

4. From the Data section, expand Sales table and drag and drop YTD Revenue field to value section.

5. From the Data section, expand Sales table and drag and drop LY YTD Revenue field to Target section



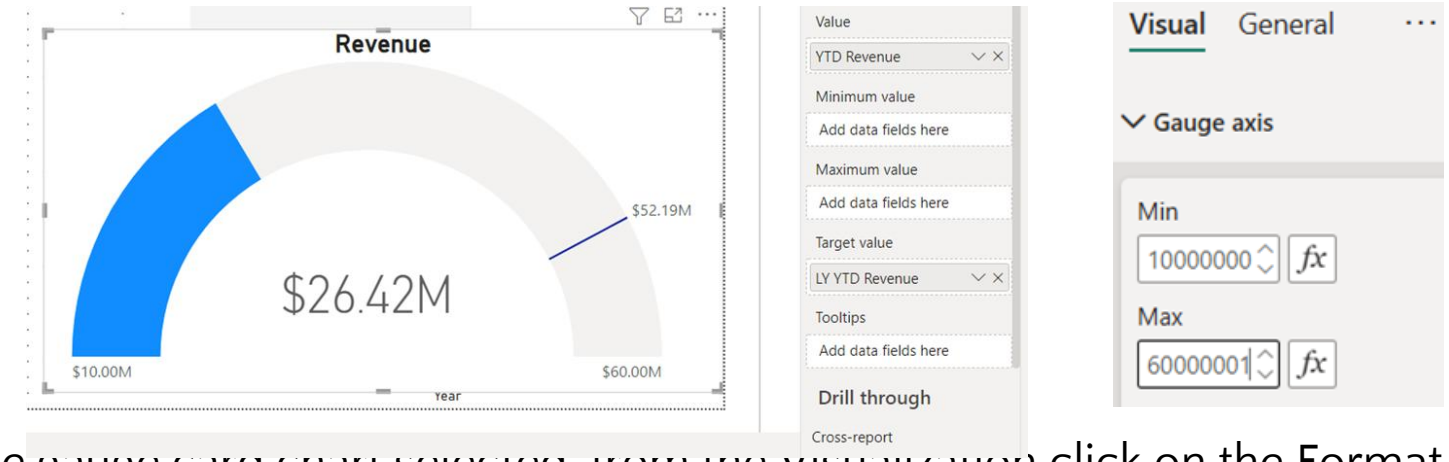


Power BI Desktop-Gauge Card

- A radial gauge chart has a circular arc and shows a single value that measures progress toward a goal or a
- Key Performance Indicator (KPI).
- The line (or needle) represents the goal or target value.
- The shading represents the progress toward that goal.
- The value inside the arc represents the progress value. Power BI spreads all possible values evenly along the arc, from the minimum (left-most value) to the maximum (right-most value)

Revenue Performance

1. From Visualizations section select Gauge Card chart.
2. From the Data section expand Sales table and drag and drop YTD Revenue to value and LY YTD revenue to Target value



5. With the gauge card chart selected, from the Visualization click on the Format icon. Modify Min and Max value

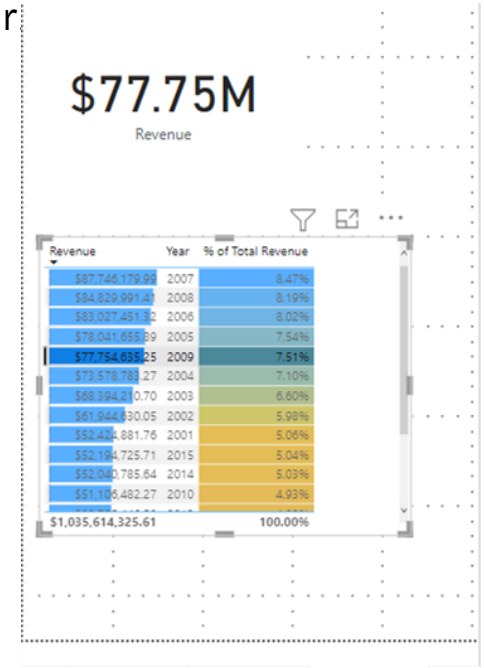
Power BI Desktop-Table

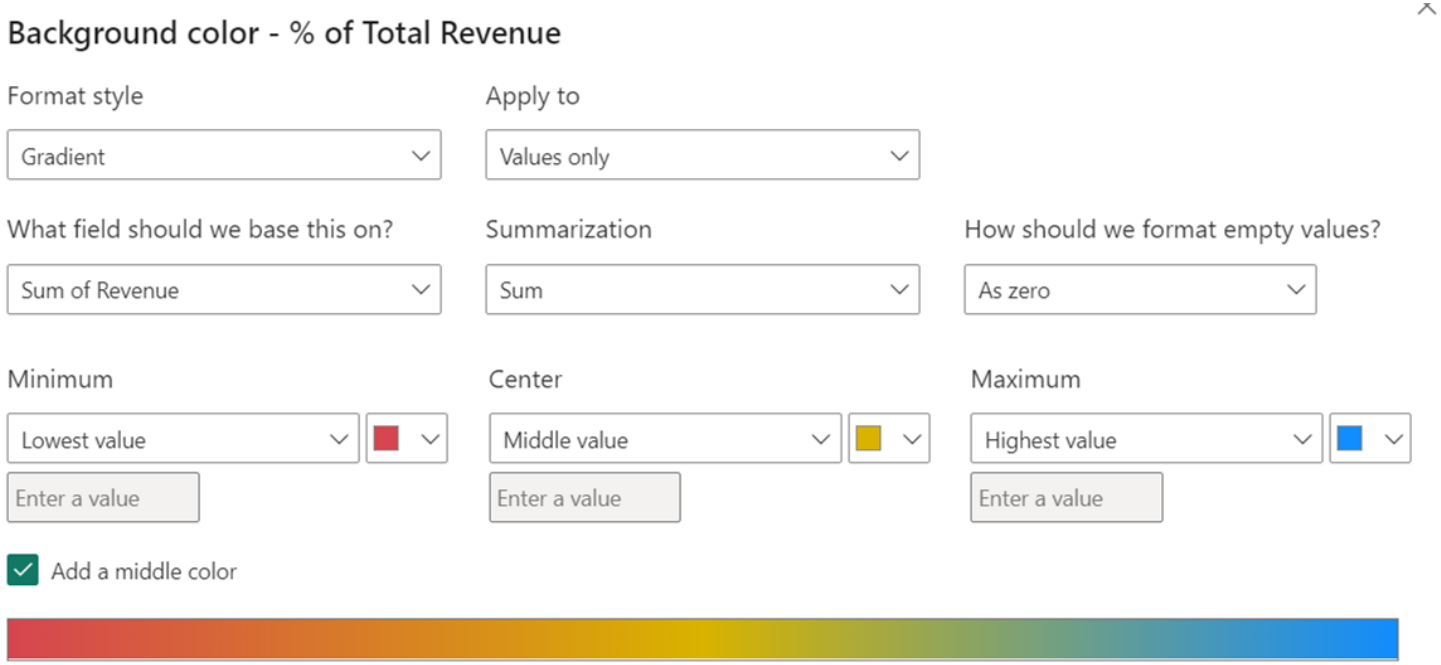
Let's assume we want to create a table visual to display the Revenue by year and add a column that displays the % of revenue for each year to help figure out the years where sales were high

1. From the Visualizations section, select Table.
2. From the Data section, and select Year. expand Date table
3. From the data section, expand Sales and select Revenue.
   Notice that Year is summed by default since the data type is of type Whole Number. We would never sum Year, so let's change it.
4. In the Data section, click on Year.
5. From the ribbon, select Column Tools -> Summarization -> Do Not Summarize.
6. From the column section of the Table visual, delete Sum of Year field by clicking on the "x".
7. From the Date table, drag Year field and drop it above Revenue in Column section of the table visual.
8. From the Column section, select the arrow next to Revenue field.
9. Select Conditional formatting -> Data bars. Data bars dialog opens. There are options to format the data bar. Select OK. Notice data bar helps to easily identify the years with high revenue and those years with low revenue.

Power BI Desktop-Table(cont.)

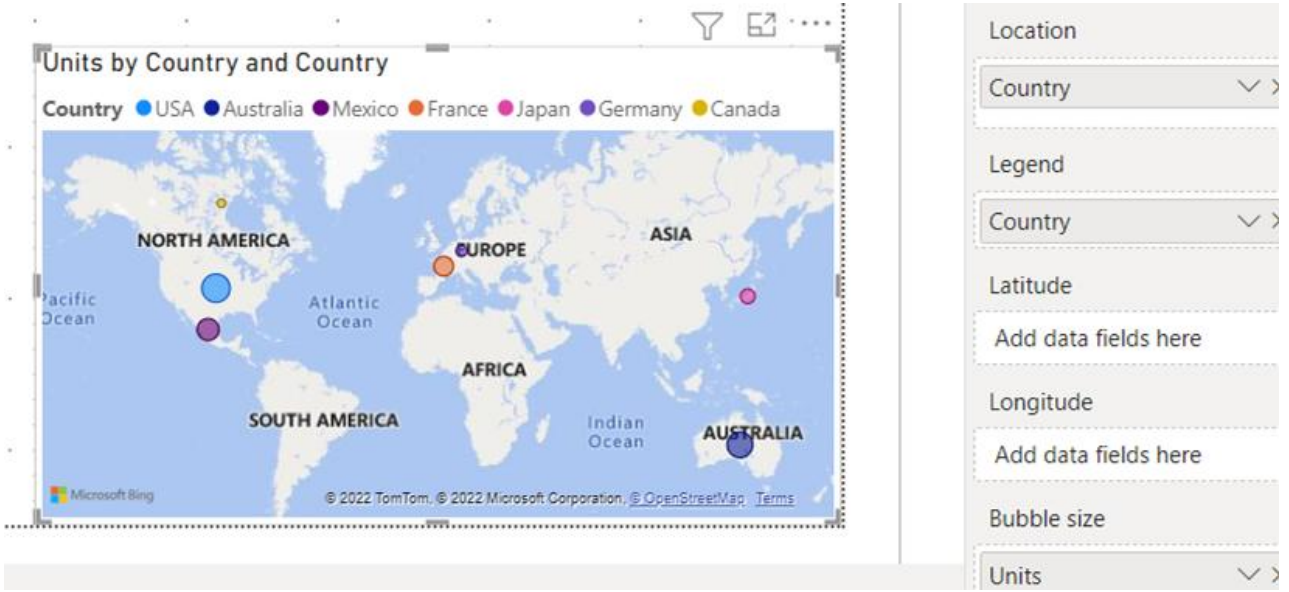
10 From the Sales table, drag Revenue field and drop it below Revenue in column section of the table visual..
11. Click the arrow next to the second Revenue field. Select Show value as -> Percent of grand total.
15. In the Column Section of the table report, click on the arrow next to % GT Revenue.. From the menu select Rename. Rename the field to % of Total Revenue.
16. Select Conditional formatting -> Background colour. Data bars dialog opens. There are options to format the data bar



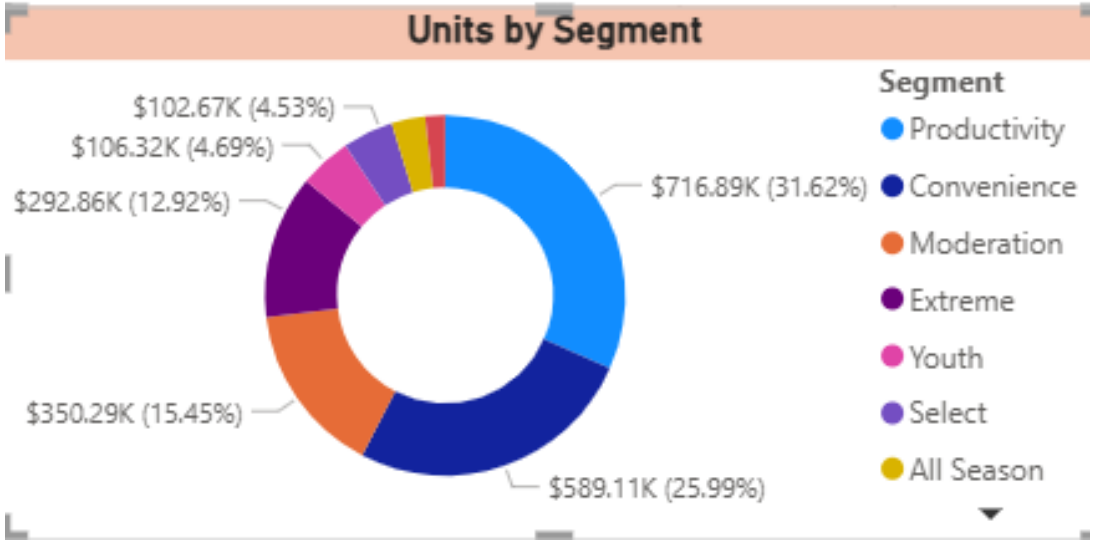


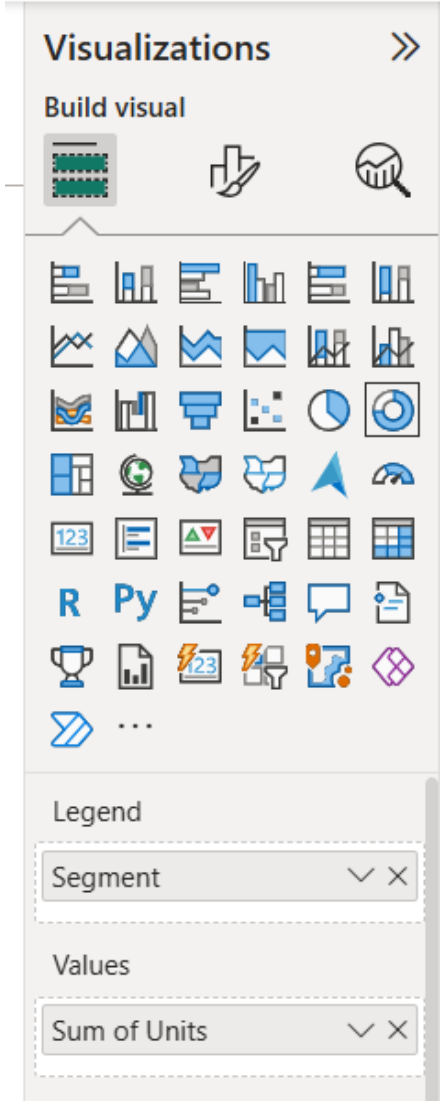
Power BI Desktop- Map

1. Click in the report canvas. Select the map visual from Visualizations.
2. From Data section, expand Geography table and drag and drop Country column to Location section of the map visual
3. From Data section, expand Sales table and drag and drop Units column to BurbleSize section of the visual.
4. From Data section, expand Geography table and drag and drop Country field to Legend section
5. Resize the map as needed. The map visual provides units sold by country



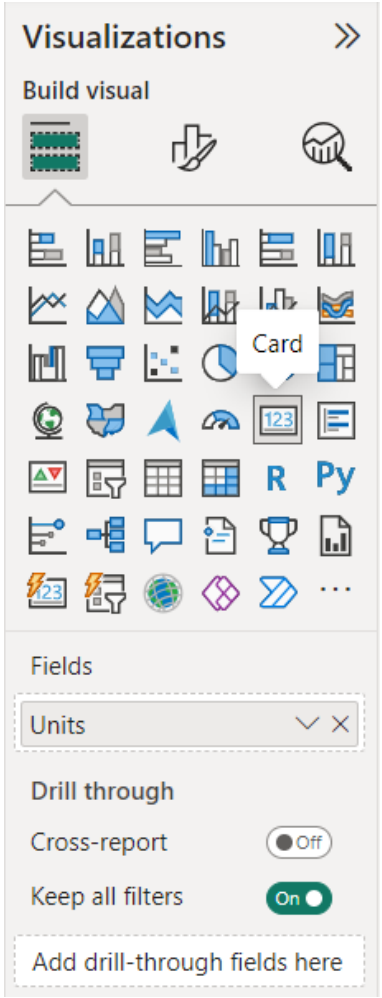
Donut Chart

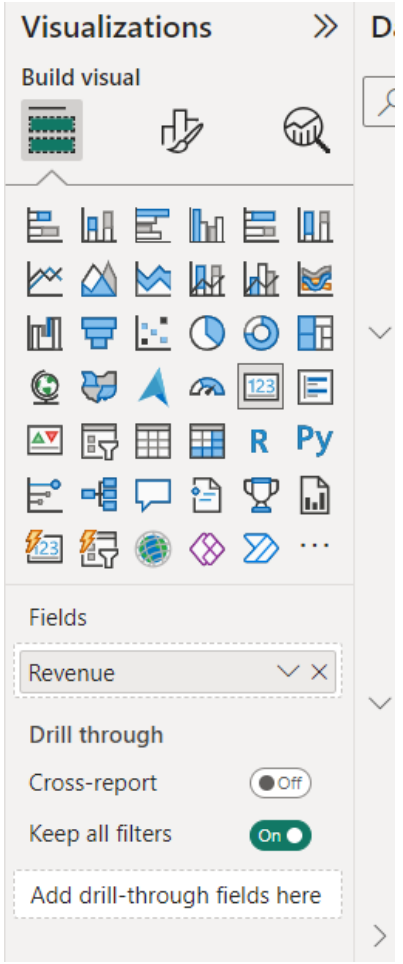

Units by Segment
$102.67K (4.53%)
$106.32K (4.69%)
$292.86K (12.92%)
$716.89K (31.62%)
$350.29K (15.45%)
$589.11K (25.99%)
Segment
Productivity
Convenience
Moderation
Extreme
Youth
Select
All Season


Visualizations
Build visual
Legend
Segment
Values
Sum of Units

Card

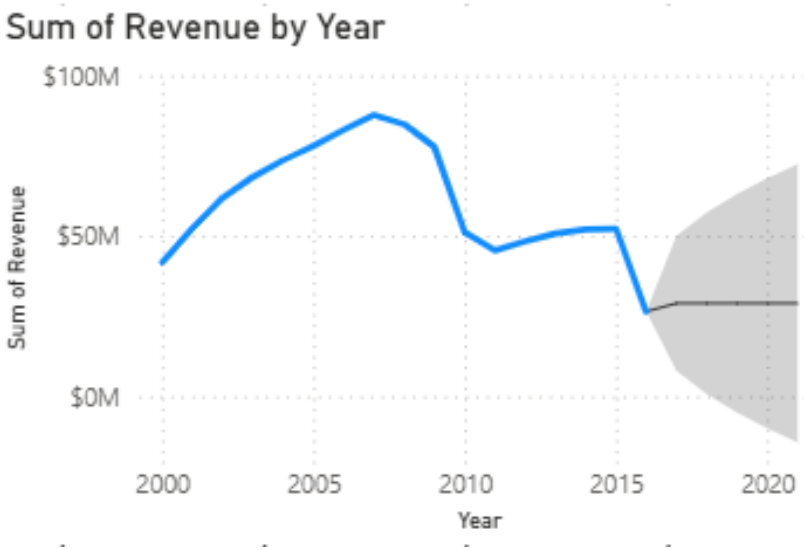
Total Revenue	Total Units
$1.04bn	$2.27M

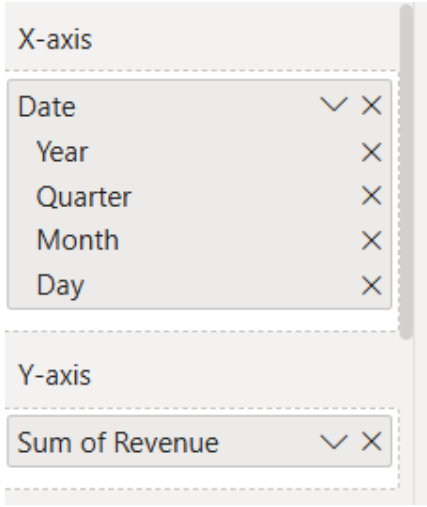


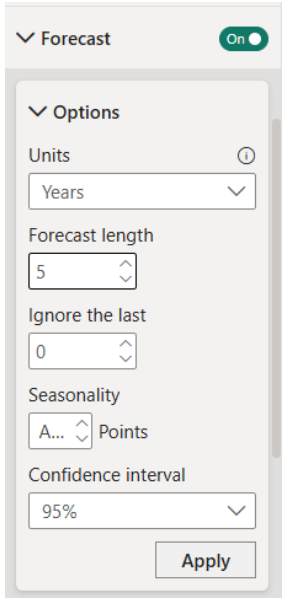


Power BI Desktop-Forecasting

Forecast next 5 years revenue.
1.Click new page report
2. From Visualizations section select Line Chart .
3. From the Data section expand Sales table and drag and drop Revenue to Y axis.
4. From the Data section expand Date table and drag and drop Date to X Axis
5. Drill up
6. With the Line chart selected, from the Visualization click on the Analytics icon.
6. Expand Forecast and select on.
7. Set the Forecast length and Confidence Interval







Power BI Desktop–Key Influencers

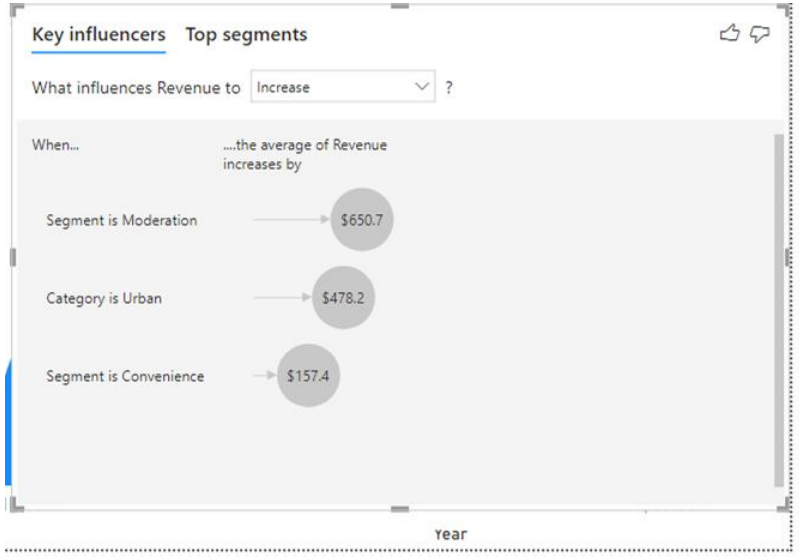
The key influencers visual helps you understand the factors that drive a metric you're interested in.

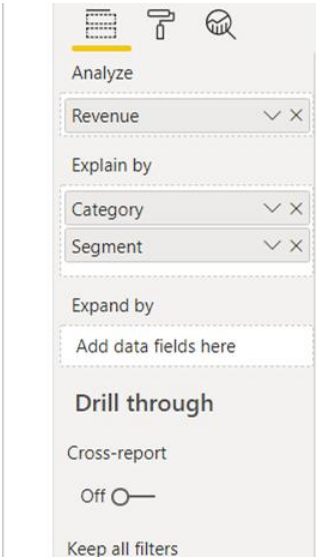
Analyse the revenue factor vs category and segment

1. From Visualizations section select Key Influencers icon .



1. From the Data section expand Sales table and drag and drop Revenue to Analyze
2. From the Data section expand Product table and drag and drop Category and Segment to Explain by value

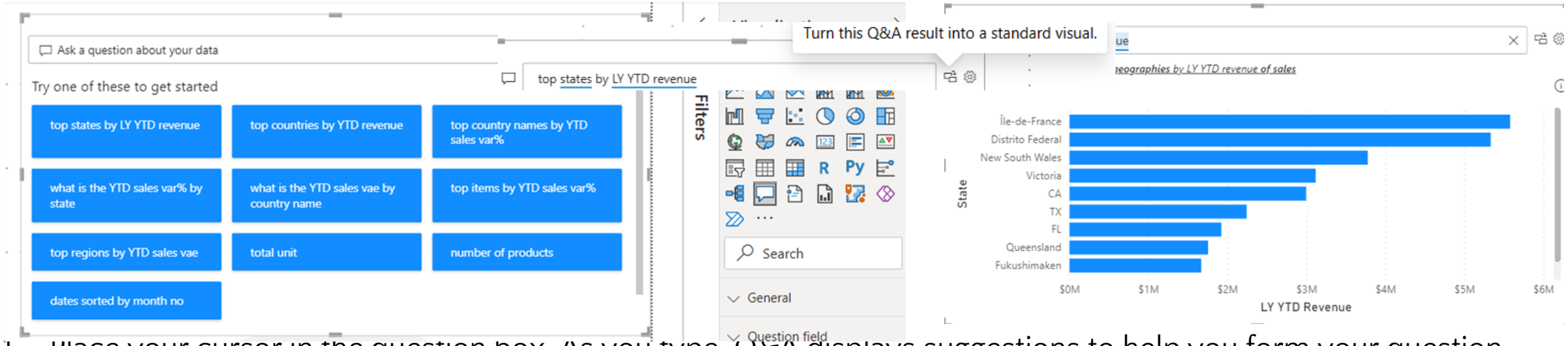




Power BI Desktop-Q & A

Use Q&A to explore your data using intuitive, natural language capabilities and receive answers in the form of charts and graphs. Q&A is different from a search engine -- Q&A only provides results about the data in Power BI datasets..

1. From Visualizations section select Q&A icon. A Q&A question box displays on your report canvas

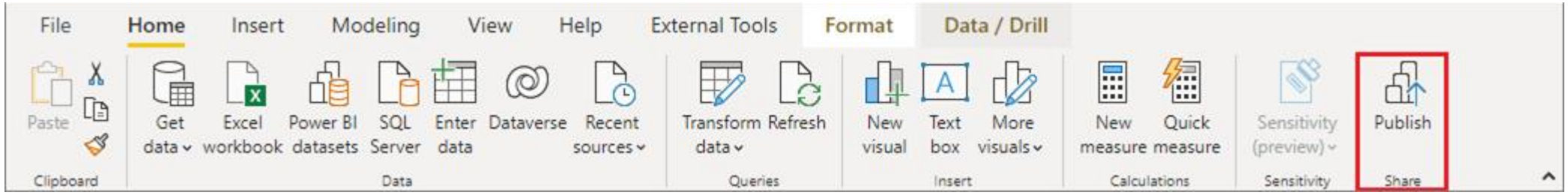


1. Place your cursor in the question box. As you type, Q&A displays suggestions to help you form your question.
2. As you type a question, Q&A picks the best visualization to display your answer.
3. Try ask a question top states bt KYTD revenue
4. Select turn this Q&A result into a standard visual

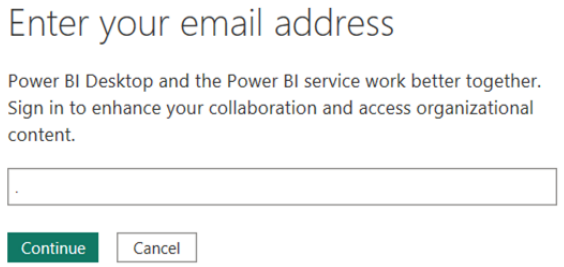
Power BI Desktop–Publish

Let's say your report in Power BI Desktop is ready, and you want other people to have access to it. One way to get it to them is to *share* it in the Power BI service. When you publish a Power BI Desktop file to the Power BI service, you publish the data in the model to your Power BI workspace or another workspace. .

1. In Power BI Desktop, choose **File** > **Publish** > **Publish to Power BI** or select **Publish** on the ribbon.



2. Sign in to Power BI, if you aren't already signed in.



Power BI requires that you use a work or school email address. You can't sign up or purchase using email addresses provided by consumer email services or telecommunication providers. This includes outlook.com, hotmail.com, gmail.com, and others.

3. If you have not signed up for a Power BI account, go to

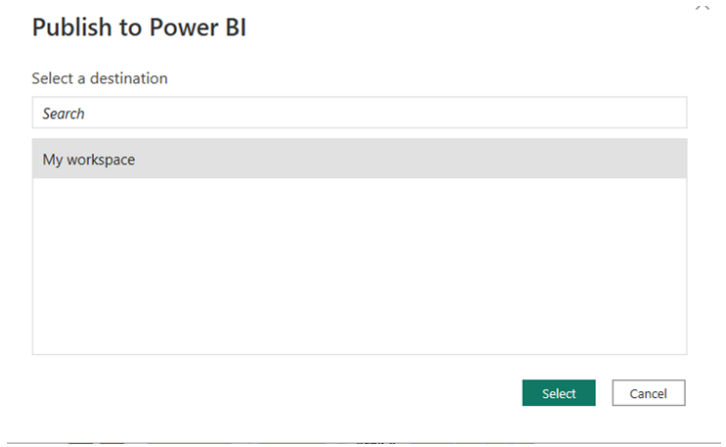
https://signup.microsoft.com/create-account/signup?sku=a403ebcc-fae0-4ca2-8c8c-7a907fd6c235&email=&ru=https:%2F%2Fapp.powerbi.com%3Fpbi_source%3Dweb%26redirectedFromSignup%3D1%26noSignUpCheck%3D1&products=a403ebcc-fae0-4ca2-8c8c-7a907fd6c235&ali=1

sign up for Power BI with a business/student email address.

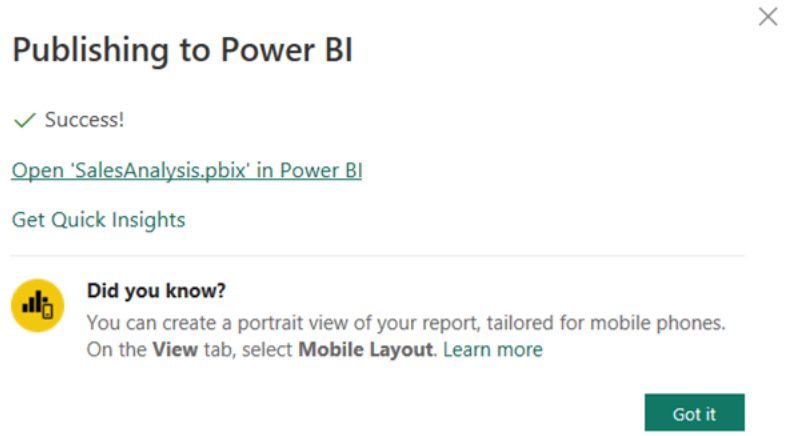
4. As you start to explore the Power BI service, you may see pop-ups asking if you want to upgrade to an individual trial of the paid version of Power BI which includes Power BI Pro and Power BI Premium Per User (PPU). If you do, select **Try now**. In some organizations, if you try to use a feature that requires a Pro or PPU license, Power BI will automatically initiate a 60 day trial of the paid version of Power BI.

Power BI Desktop–Publish

3. Select the destination. You can search your list of available workspaces to find the workspace into which you want to publish. Select the workspace, and then click the **Select** button to publish.



4. When publishing is complete, you receive a link to your report. Select the link to open the report in your Power BI site. You will be directly navigated to your Workspace



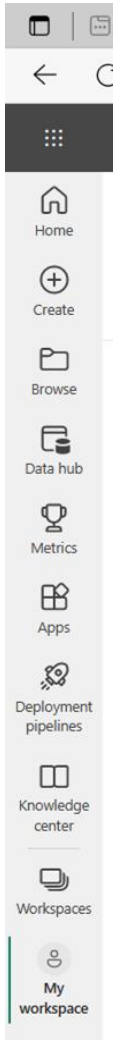
Workspaces

Workspaces are created on capacities. Essentially, they are containers for dashboards, reports, workbooks, datasets, and dataflows in Power BI.

There are two types of workspaces: *My workspace* and *workspaces*.

- *My workspace* is the personal workspace for any Power BI customer to work with your own content. Only you have access to your My workspace. You can share dashboards and reports from your My Workspace. If you want to collaborate on dashboards and reports, or create an app, then you want to work in a workspace.

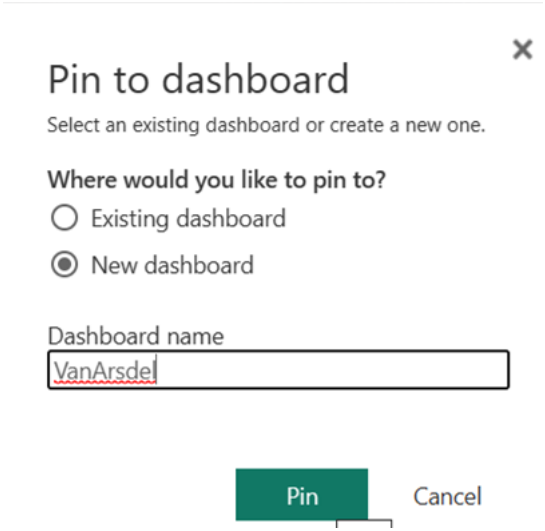
- *Workspaces* are used to collaborate and share content with colleagues. You can add colleagues to your workspaces and collaborate on dashboards, reports, workbooks, and datasets. With one exception, each workspace member needs a Power BI Pro or Premium Per User (PPU) license.



Power BI Service–Dashboard

1. To create an operational dashboard, you need to open the report and select visuals that provide metrics for the organization.
2. Hover over Revenue by Year and Manufacturer line chart. A pin icon will appear on the top right corner of the visual. The pin icon will appear for each visual you would like to pin to the dashboard.
3. Click on the Pin icon. This brings up a dialog box. Enter a name for the dashboard you want to pin to, then click or tap **Pin**..





Power BI Service-Dashboad

4. Hover over Revenue KPI visual and click on the Pin icon and pin it to VanArsdel dashboard.

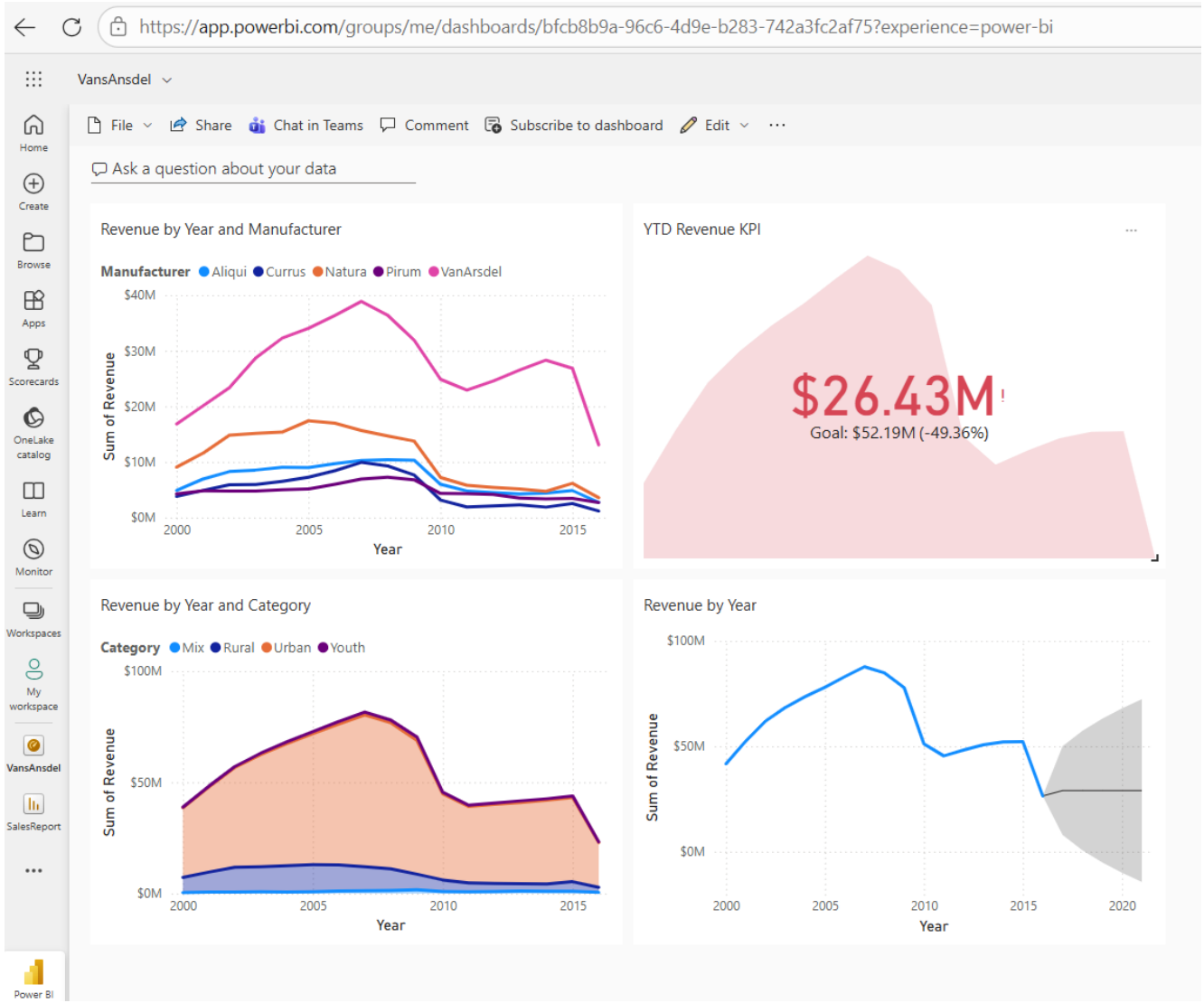
7. Hover over by Year and Category stacked area chart and click on the Pin icon and pin it to VanArsdel dashboard.

8. Hover over Units by Country chart and click on the Pin icon and pin it to VanArsdel dashboard.

9. Hover over revenue by year forecast chart and click on the Pin icon and pin it to VanArs.del dashboard.

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Dashboard


https://app.powerbi.com/groups/me/dashboards/bfcb8b9a-96c6-4d9e-b283-742a3fc2af75?experience=power-bi
VansAnsdel
File
Share
Chat in Teams
Comment
Subscribe to dashboard
Edit
Ask a question about your data
Home
Create
Browse
Apps
Scorecards
OneLake catalog
Learn
Monitor
Workspaces
My workspace
VansAnsdel
SalesReport
Power BI
Revenue by Year and Manufacturer
Manufacturer Aliqui Currus Natura Pirum VanArsdel
$40M
$30M
$20M
$10M
$0M
Sum of Revenue
2000
2005
2010
2015
Year
YTD Revenue KPI
$26.43M
Goal: $52.19M (-49.36%)
Revenue by Year and Category
Category Mix Rural Urban Youth
$100M
$50M
$0M
Sum of Revenue
2000
2005
2010
2015
Year
Revenue by Year
$100M
$50M
$0M
Sum of Revenue
2000
2005
2010
2015
2020
Year